Comparison of Tesseract OCR, Easy OCR, and Transformer OCR on Handwritten Image

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***Abstract*—This study compares the performance of Tesseract, Easy-OCR, and Transformer OCR in recognizing crossed-out text in the Indonesian and English languages. The focus on crossed-out text aims to assess the ability of OCR methods to face the challenges of recognizing text with non-standard formats. This preliminary study provides an important finding for optimizing the OCR technology in crossed-out multilingual handwriting. Testing was conducted on the modified IAM Handwriting dataset and the UM-PTI-Handwriting dataset using the processing time, Character Error Rate (CER), and Word Error Rate (WER) metrics to compare Transformer (Tr-OCR, Donut), Tesseract, and Easy OCR. According to the WER and CER metrics, Transformer-based OCR (Tr-OCR) achieves the lowest error rate, achieving 0,63 WER and 0.63 CER on the modified IAM Handwriting dataset, while 1,92 WER and 3,05 CER on the UM-PTI-Handwriting dataset. In terms of the processing time, Tesseract OCR is the fastest, while Transformer-based OCR (Donut) is the slowest. It concluded that the Transformer OCR (Tr-OCR) excels in recognizing English handwritten. However, all compared OCRs have lower accuracy in Indonesian handwritten text compared to English handwritten text.**

Keywords—Transformer OCR, Easy-OCR, Tesseract OCR, Multilingual Handwriting, Crossed-Out.

I. Introduction

Handwriting activities have been empirically proven to influence the acquisition of alphabetic and orthographic knowledge, showing the word form in early childhood [1]. Graphomotor actions that represent the physical movement of writing and the variability of produced letter shapes play a role in supporting literacy deployment [2]. Additionally, self-regulated learning in writing shows positive changes in cognitive techniques [3], [4]. Handwriting can also detect learning disorders in children early [5]–[7]. The major role of writing activities has sparked the development of handwriting-based examination technology as a support in education. Recognizing the crossed-out handwritten answers using Optical Character Recognition (OCR) is the most challenging part of writing activities in big data[8]–[10].

Several previous studies have observed the use of OCR. Transformer, Tesseract, and Easy OCR have a more promising performance than the comparison OCRs. Jain et al. (2021) show that Tesseract OCR outperforms Ocrad, GoCR, and OCRopus for computer-written based on multilingual, operating system, and file support criteria [11]. Tarun Kumar et al. (2025) showed that Easy OCR is more effective for higher resolution images (64x), while Encoder and Decoder OCR are more effective for lower resolution images (28x) [12]. Whereas W. Khallouli et al. (2024) show that Transformer OCR is more robust than Keras OCR [13].

Several studies have measured the accuracy of Tesseract OCR as a technology developed using a rule-based and machine learning approach [14]. In the education field, the modified Tesseract OCR can detect student names on the apprenticeship acceptance letter, and only produces an average error of 9% based on internship eligibility [15]. Tesseract OCR also excels in recognizing computer-written and handwritten [16]. However, Tesseract OCR fails to detect similar letters and numbers, such as S and 5 [8], [17], [18]. Chinta et al. (2024) applied Tesseract OCR to extract the invoice information, resulting in high accuracy [19]. On the contrary, the use of Tesseract OCR resulted in the lowest accuracy in medicine handwritten compared to Yolo [20]. Moreover, Joshi (2024) observed that the use of Tesseract OCR has challenges in low-resolution and noisy images [18]. No specific research has evaluated Tesseract OCR on multilingual crossed-out handwriting, and therefore, further observation is needed.

An Easy OCR, which applies a convolutional neural network (CNN) as a basic framework [21], [22]. Easy OCR excels in the noise of Bangla handwritten, such as blurring, tearing, and skewing [21]. Pattanayak et al. (2023) modified Easy OCR to recognize English handwritten text by adding a set of image patches[23]. The combination of Yolo and Easy OCR to detect Russian and Indian text achieves an accuracy of 0.9 [24]. No specific research has evaluated Easy OCR on multilingual crossed-out handwriting, and therefore, further observation is needed.

Transformer OCR is an encoder and decoder approach [25]. On a dataset of 1.3 million annotated Persian texts, Transformer OCR achieves 77.25% accuracy against a CNN model with 3.21 times faster training [26]. The combination of Yolov4 and Transformer OCR to recognize Vietnamese writing, with an accuracy of 84.06%, with a word error rate reaching 22.67%[27]. In contrast, Dipu (2021) showed that the performance of the OCR transformer is worse compared to the state-of-the-art Bangla OCR[28]. The conflicting results in the Transformer OCR study require further study.

Although many studies have compared OCR, there is a gap in comprehensive analyses of the recognition of crossed-out text in Indonesian and English documents, particularly regarding the comparison of Tesseract, Easy-OCR, and Transformer-based OCR in terms of their robustness. This preliminary research fills this gap by conducting a comprehensive evaluation in terms of character and word accuracy, as well as processing time efficiency for finding the optimized OCR in crossed-out multilingual handwriting. The main contribution of this research is to provide a systematic empirical evaluation of the advantages and limitations of each OCR method in the context of multilingual and crossed-out text.

II. METHODS

This study aims to compare the performance of several OCR methods on crossed-out handwritten text in English and Indonesian. This section outlines the dataset preparation and configuration, as well as the comparison and evaluation of OCR methods, including Transformer (Donut, Tr-OCR), Tesseract, and Easy OCR, as depicted in Fig.1.

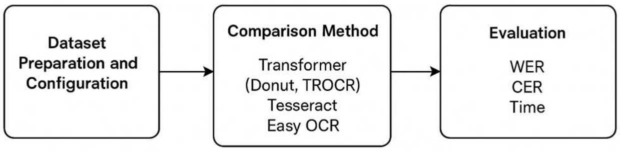

Fig. 1. Research Methodology

A. Dataset Preparation and Configuration

We use two datasets: the modified IAM Handwriting - Line-level Database and the UM-PTI-Handwriting. The IAM Handwriting-Line-level Database is obtained from the Hugging Face platform [29]. This dataset is part of the IAM Handwriting Database, which contains handwritten line snippets in English from various authors. The instant data consists of an image and text showing the image's transcription. In this study, 200 handwriting-line images were randomly added with 5 to 50 synthetic strikethroughs. Crossed-out handwriting is categorized based on the level of readability, which is divided into two categories: 100 massive crossed-out handwriting and 100 readable crossed-out handwriting, as shown in Fig. 2. The test of the IAM Handwriting Database aims to compare the OCRs on crossed-out English handwriting.

Testing was also conducted on the UM-PTI-Handwriting Dataset [30] to assess the effectiveness of each method in recognizing crossed-out Indonesian handwriting. This dataset contains 20 images of inline text, showing the answers of eight UM informatics engineering students on the Database Mid-Term Exam. Each image is scanned at a resolution of 200 DPI. An example of UM-PTI-Handwriting is shown in Fig. 3. Both datasets were annotated and labelled using Roboflow. The modified image transcription is validated based on the validator's readability [1].

B. Tesseract OCR

Tesseract is an open-source OCR that traditionally involves a multi-stage process relying on feature extraction and pattern matching. Based on Fig. 4, the Tesseract framework consists of: image preprocessing, feature extraction, character recognition, and post-processing [31].

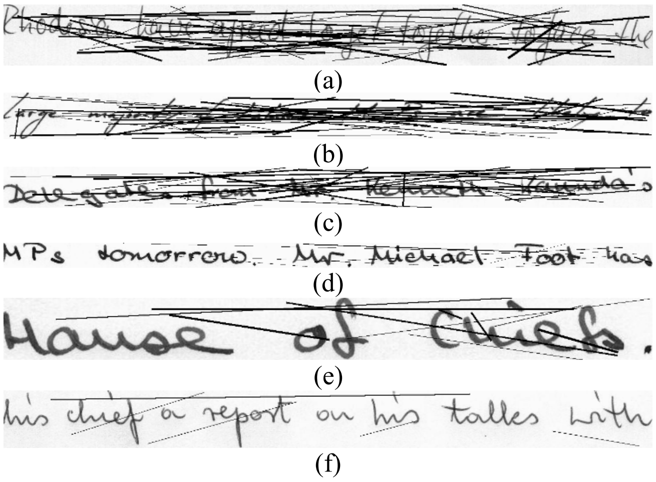

Fig. 2. The Modified IAM Handwriting - Line-Level Database (a)-(c) Massive Crossed-Out (d)-(f) Readable Images With Crossed-out

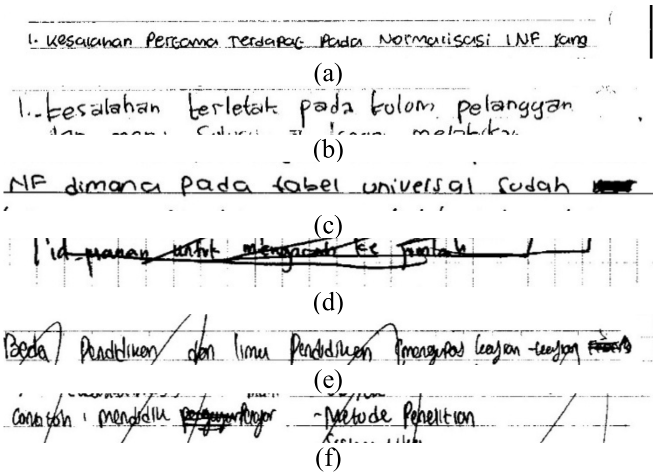

Fig. 3. Sample of UM-PTI-Handwriting (a)-(c) Unstroke-out Handwriting (d)-(f) Crossed-Out Handwriting

Fig. 4. Tesseract Framework [15]

In Image Preprocessing, the input image undergoes several pre-processing steps, including: denoising, binarization, skew correction, line and word segmentation, and character segmentation [32]. The next stage is connected handwriting, which is feature extraction. Once characters are segmented, Tesseract extracts relevant features from each character image. These structural features can include loops, corners, line segments, and statistical features based on pixel patterns. The third stage is character recognition, which compares against a database of character patterns learned during the training phase. This matching process aims to identify the character most likely to correspond to the extracted features. Post-processing is the last stage includes dictionary lookups and language model integration.

C. Easy OCR

Easy OCR simplifies a Convolutional Neural Network (CNN)-based model. Based on Fig.5, Easy OCR employs a CRAFT (Character Region Awareness for Text detection) to identify and locate text regions within the input image[31].

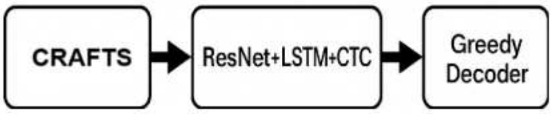

Fig. 5. EasyOCR Framework [32]

CRAFTS resulted in the bounding boxes around detected text. In the middle process, LSTM is used to sort features from text images, while CTC is used to train a model without label segmentation per character.

D. Transformer OCR

Transformer OCR represents a new approach that leverages the encoder and decoder architecture shown in Fig. 6. The Transformer encoder extracts visual features based on position and patch encoding to learns contextual relationships between different parts of the visual feature map, allowing the model to understand the spatial arrangement and the visual context. The self-attention decoder is the core of the Transformer OCR. It autoregressively generates the text sequence. At each generation, an attention mechanism allows the decoder to focus on the relevant patch of the encoded visual features. This attention mechanism enables the model to align the generated characters with the corresponding visual information in the input image, even for distorted or irregularly shaped handwriting. The decoder generates characters until an end-of-sequence token is predicted, resulting in the final transcribed text in the output sequence generation. This study employs two Transformer models: the Tr-OCR (pre-trained Microsoft/trocr-base-handwritten) that outputs plain text, and the Donut that outputs seq-to-JSON.

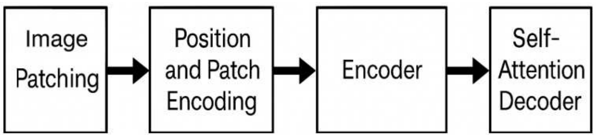

Fig. 6. Transformer Framework

E. Evaluation Environment

The system environment used in this study employed an Intel Xeon processor with a speed of 2.20 GHz, 2 cores, and supports hyper-threading technology with 2 threads per core. The available main memory has a capacity of 12 GB of RAM. Experiments were performed in a fully virtualized environment using the KVM hypervisor. There was no GPU device. The operating system used was Linux with kernel version 6.1.123+ running on a 64-bit architecture.

F. Comparing Evaluation

Testing was conducted on the modified IAM Handwriting dataset and the UM-PTI-Handwriting dataset using the processing time, Character Error Rate (CER), and Word Error Rate (WER) metrics to compare Transformer (Tr-OCR, Donut), Tesseract, and Easy OCR. Testing time is measured in seconds (s). WER and CER are used to assess the accuracy of OCR and show the variability between languages and text types. WER quantifies the number of errors made at the word level, which is shown in Equation (1), whereas CER quantifies the number of errors at the character level, which is shown in Equation (2).

$$WER = \frac{Sw+Iw+Dw}{Nw}, \tag{1}$$

$$CER = \frac{Sc+Ic+Dc}{Nc}, \tag{2}$$

where S is the total incorrect *words* w or characters c replaced by the correct pattern. D is the total of correct words or characters missing from the prediction. I is the total of extra incorrect words or characters present in the prediction.

III. Results and Discussion

In this study, we provide a systematic empirical evaluation of Transformer OCR, Tesseract OCR, and Easy OCR, which is shown in Table I. Based on Table I, it can be concluded that Tesseract OCR is the fastest but has the highest WER and CER, while Transformer OCR (Donut) is the slowest and more sensitive to the cross-out-text. Transformer OCR (Tr-OCR) excels in recognizing English handwritten text compared to Tesseract and Easy OCR, but fails to detect Multi-Line Words, as shown in Fig. 7.

The comparison of the crossed-out English handwriting recognition in the Modified IAM Handwriting testing is shown in Table II. Tesseract OCR and Easy OCR produce poor recognition at massive and non-massive crossed-out text. It is supported by the word error rate boxplot shown in Fig. 9–10. Transformer OCR (Tr-OCR) produces better performance. However, the text recognition results on the readable image with crossed-out text are better than those on massive crossed-out text. Crossed-out characters cause misperception of letters, so OCR predicts the failure words. Massive scribbles in some texts cause some of the writing to be blurred. For example, in Fig. 2(a), the massively crossed-out location starts from the "sia" word, so the Transformer is only able to recognize the "Rodhes" from "Rodhesia" word shown in Table I(a). In Table 1(d), the strikethrough causes the word merging, such as MPs instead of M Ps.

The comparison of methods is also illustrated using the box plot. The Y-axis shows the evaluation metrics. Each box shows the distribution of Quartiles 1 to 3. The box (whiskers) indicates the range of non-outlier data, while data outside the whiskers are considered outliers.

Fig. 8 is a boxplot showing the time distribution of Transformer OCR (Tr-OCR and Donut), Tesseract OCR, and Easy OCR. Transformer OCR (Donut) produces the longest processing time, with the median around 67–70 seconds. The computing time of Easy OCR resembles the Transformer OCR (Tr-OCR) , with a median of around 7–8 seconds. The time computed by Tesseract OCR reaches the fastest time.

Fig. 7. Failure of Transformer OCR to Detect Multi-Line Words

TABLE I. Comparison of The Results

Testing	*Dataset*	*Result*			
		Transformer OCR		*Tesseract OCR*	*Easy OCR*
		Tr-OCR	*Donut*		
Time	1	8,04	63,58	0,24	7,12
	2	12,19	98,27	0,40	5,09
WER	1	0,63	0,97	1,06	1,00
	2	1,92	1,97	2,24	2,22
CER	1	0,63	0,94	0,96	0,96
	2	3,05	3,05	5,48	4,39

[1] The Modified IAM Handwriting
[2] UM-PTI-Handwriting

TABLE II. COMPARISON OF TRANSFORMER, TESSERACT, AND EASY OCR IN THE MODIFIED IAM HANDWRITING - LINE-LEVEL DATABASE

ID Fig.2	Reference	Result			
		Transformer OCR		*Tesseract OCR*	*Easy OCR*
		Tr-OCR	*Donut*		
(a)	Rhodesia have agreed to get together to face the	Rhodes have been experienced by those figures of the	the		
(b)	large majority of Labour M Ps are likely to	large , may be extended to a table to the	the cat		
(c)	Delegates from Mr. Kenneth Kaunda's	Although other French lawyers were	independent		"ae 2 Jts
(d)	M Ps tomorrow. Mr. Michael Foot has	MPs tomorrow. Mr. Michael Foot has	mrs tomorrow	MPs	4 Ps donıorf eio 11 1texae =F2 1oza
(e)	House of Chiefs .	House of their life .	harris		44t2
(f)	his chief a report on his talks with	his chief accept on his talks with	this chief or report on his tables with		u2 OC 1b0 Ou h Fle...

Fig. 9 is a box plot of the compared OCRs in the Modified IAM Handwriting testing. The lowers WER value indicates an accurate recognition result. The WER distribution of Tr-OCR is the lowest, with a median around 0.65. The median WER of Donut, Tesseract, and Easy-OCR is around 1.0, showing the worst accuracy.

Fig. 10 shows the CER of the compared OCRs in the Modified IAM Handwriting testing. The lower WER value indicates an accurate recognition result. The WER distribution of Tr-OCR is the lowest, with a median around 0.65. The median CER of Donut, Tesseract, and Easy-OCR is around 1.0, showing the worst accuracy. The presence of many outliers produced by the third OCR suggests instability when recognizing massive crossed-out text. Based on Fig. 9–10, Tr-OCR (Transformer) produces the most reliable results in the cross-out English field, as indicated by the lowest CER and WER.

The comparison of the crossed-out Indonesian handwriting recognition in the UM-PTI-Handwriting testing is shown in Table III. Tesseract OCR and Easy OCR also produce poor recognition at massive and non-massive crossed-out text. It is supported by the word error rate boxplot shown in Fig. 12–13. Transformer OCR (Tr-OCR) produces better performance. The Indonesian recognition of uncrossed-out text is better than the crossed-out text. Crossed-out characters cause misperception of letters. For example, the “id_psanan” word in Fig. 3(c) and the “beda” word in Fig. 3(d) that included strokes are recognized by the fourth OCR with the different words, illustrated in Table III. However, the prediction of Tr-OCR is closer to the reference, such as the "contoh" word in Fig.3(f), which is predicted as “canyon”.

Fig. 11 is a box plot showing the time distribution of Transformer OCR, Tesseract OCR, and Easy OCR in the UM-PTI-Handwriting testing. Regarding the timing test on the Modified IAM Handwriting dataset, Tesseract OCR produces the best timing, while Donut is the slowest OCR.

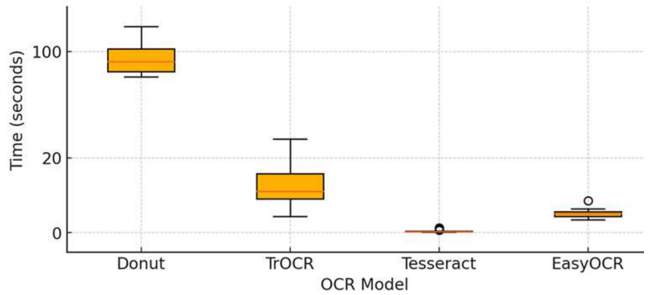

Fig. 8. Time Comparison of OCRs on the modified IAM Database

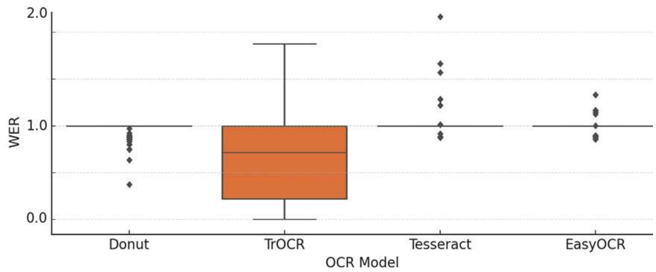

Fig. 9. WER Comparison of OCRs on the modified IAM Database

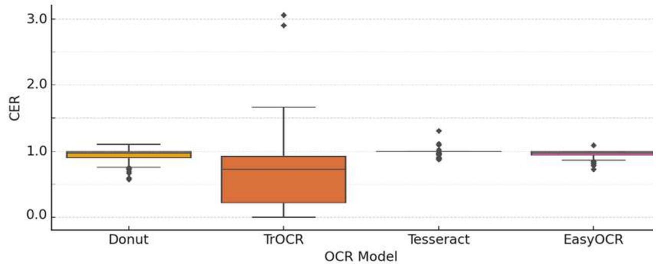

Fig. 10. *CER (*Comparison of OCRs on the Modified IAM Database

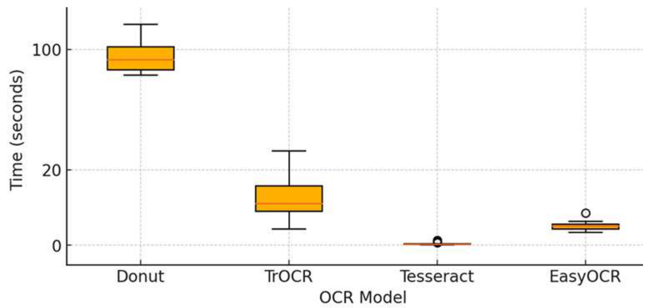

Fig. 11. Time Comparison of OCRs on the UM-PTI-Handwriting Database

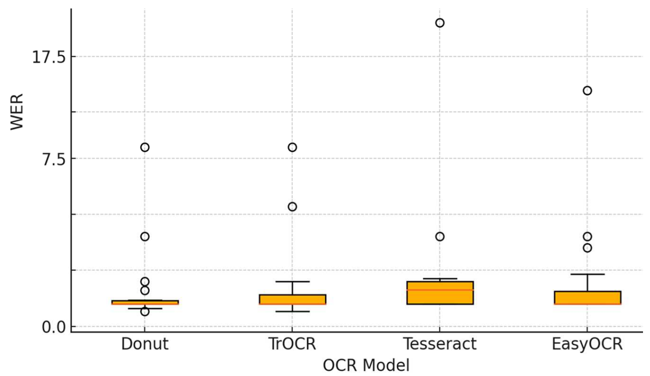

Fig. 12. *WER* Comparison of OCRs in the UM-PTI-Handwriting Database

TABLE III. RESULT COMPARISON OF TRANSFORMER OCR, TESSERACT OCR, AND EASY OCR IN THE UM-PTI-HANDWRITING DATABASE

Id Fig. 3	Reference	Result			
		Transformer OCR		***Tesseract OCR***	***Easy OCR***
		Tr-OCR	***Donut***		
(a)	1. kesalahan pertama terdapat pada normalisasi 1NF yang	" Kesakhan Personal resource Rada Nkomalisasi INE rang , "	i. fesalahan	i acai 2 ic i RI -~ { \. Ke@Saianan_ Pestcwnen 1 Meter Pa Practica. Notmentisesi.	vegaicabaan_ Perccmnca IerdaBC Prsdca NormquiscsSi LME Yaına
(b)	1. kesalahan terletak pada kolom pelanggan	1 . Kesahahn terletat pads kulom pelangyan	densbleon	\besaleban erletak peda bolom pelanggar 1 -' ¥ TE ey ems a Sc coy emi Lael	1Fesalahan [erlełal Folom Pelangyan PaJa
(c)	NF dimana pada tabel universal sudah	NF dimona pada label universal Judah 8	nine dimance bada	NE dimanca Pada 4oabet univelSG\ Codah mmr	dimansı Pada {alel LhiVeLIaL Codah NE
(d)	Id_psanan untuk mengarah ke	first-programming with the undergraduate practice of the	intractional		14-tanew <Huk #=2 E+
(e)	Beda Pendidikan dan Ilmu Pendidikan mengupas	Robert Purdelman , don't now . Purdidough-Imangued began-wagon	perbidman	eK) Tact” ln luna Pecbticag” Voorn) lnjon cpa xt	@od pndblkoy @nadnyn Jooagpo) Lajm _teJp] 24s lımu
(f)	Contoh mendidik - metode penelitian	canyon , metaphor stronger - reliance Robertson - ?	metalde persell ton		#h Maddn FFzafeh Aulde_Pehell ton @lb

Fig. 12 shows the WER of the compared OCRs in the UM-PTI-Handwriting testing. The whiskers' position in Fig. 12 indicated that Tr-OCR and Donut fared similarly, but Easy OCR had a higher WER. Tesseract OCR, which has the highest WER, shows the most difficulty in recognizing Indonesian strikethrough words. It is caused by pre-training primarily using English-language documents.

Fig. 13 shows the CER of the compared OCRs in the UM-PTI-Handwriting testing, respectively. The comparison of the box patterns in Fig. 13, which are similar to Fig. 12, indicates that the CER of the OCRs pattern is worse than the WER pattern. However, the character accuracy is worse. It was also emphasized in Table II that all OCRs reached a high CER of above 3. In the crossed-out handwritten Indonesian, the WER produced by the compared methods is high, indicating that the model needs fine-tuning on an Indonesian dataset specifically for the Indonesian language.

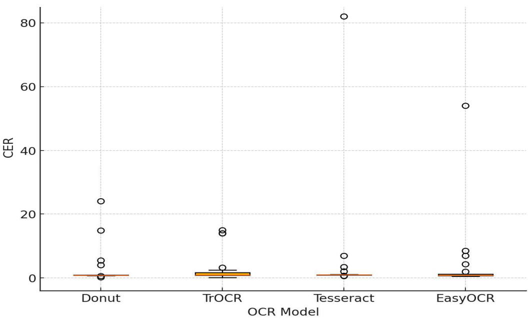

Fig. 13. *CER Comparison* of OCRs on the UM-PTI-Handwriting Dataset

According to Fig. 12–13, a few outliers resulted in compared OCRs requiring further feature optimization. The stroke on the handwritten disturbed the low-level feature, indicating the need to remove the strikethroughs. On the other hand, a strikethrough signifies the deletion of a misspelled word. Therefore, it is necessary to recognize the pattern of crossed-out text and the patterns that do not indicate deletion.

IV. CONCLUSION

We concluded that the Transformer OCR (Tr-OCR) excels in recognizing English handwritten text than Tesseract and Easy OCR. Tr-OCR produces WER and CER higher than Donut, Tesseract, and Easy-OCR for English and Indonesian handwriting. However, all OCRs showed lower accuracy when recognizing Indonesian handwritten text compared to English handwritten text. Donut performed the slowest, while Tesseract OCR consistently performed the fastest. Crossed-out interference often obscures the text that is being detected. In the context of evaluation, a deep understanding of the type of scribbling is necessary to optimize, allowing OCR to be more selective in detecting words. Whereas, all three transformers produced poor accuracy in recognizing Indonesian handwriting.

Future research needs to consider the detection of handwriting strikethrough based on the type of strikethrough, multi-line text recognition, and fine-tune for the Indonesian handwritten dataset.

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